Domain Fusion Contrastive Learning for Cross-Scene Hyperspectral Image Classification

Zhao Qiu, Jie Xu, Jiangtao Peng, *Senior Member, IEEE*, and Weiwei Sun, *Senior Member, IEEE*

***Abstract*— Recently, domain adaptation (DA) methods based on contrastive learning are widely used to solve the cross-scene classification problem. However, existing contrastive learning methods only focus on source domain or target domain features, or do not adequately consider the interaction of domain information, thus the learned domain-invariant features still have large discrepancies. To address this problem, we propose a novel domain fusion contrastive learning (DFCL) framework for cross-scene hyperspectral image (HSI) classification. DFCL uses an interdomain and intradomain dual-domain fusion strategy at the feature level, which introduces domain information as a noise interference term for sample enhancement. With the interference of domain information, same category samples are pulled closer and different categories samples are pushed further apart to learn more discriminative features. In addition, we construct an intermediate domain through the source and target domains and define a feature space loss that measures domain discrepancy by feature similarity and label similarity. Finally, a progressive selection strategy based on prototype learning is proposed to select high-confidence pseudolabels for DFCL. Experiments on three HSI cross-scene datasets show that the proposed method is superior to existing DA methods.**

***Index Terms*— Contrastive learning, domain adaptation (DA), domain fusion, hyperspectral image (HSI) classification.**

I. Introduction

With the development of various image processing technologies and advanced sensors, hyperspectral images (HSIs) are widely used in various fields, such as vegetation ecology [1] and mineral exploration [2]. HSIs are different from traditional optical images. They contain hundreds of bands and rich spectral information, showing different fine spectral features for different substances [3]. The rich information contained is often used to classify surface objects, classify each pixel according to its spectral characteristics, and use the obtained classification map to further study land cover distribution [4], [5]. The existing HSI classification methods require a large number of labeled samples for training and learning their features, so as to achieve better classification results. However, researchers have gradually realized that obtaining sample labels became a major problem, which usually takes a lot of time and manpower to process. Limited sample labels will limit the HSI classification effect [6].

In the past few decades, many researchers have considered using models learned from the existing HSIs to classify HSIs with the same category. Unfortunately, different HSIs have spectral shifts due to different sensors, surface variation, and different sampling times [7]. Therefore, this method usually fails to achieve satisfactory results. Fortunately, domain adaptation (DA) technology can solve the spectral shift problem well without the need for new labeled samples [8], [9]. Traditional DA methods generally align the distribution of the source domain and the target domain, such as subspace alignment [10], correlation alignment (CORAL) [11], and discriminative cooperative alignment (DCA) [12]. However, traditional DA methods rely on handcrafted features, resulting in models that are limited in their ability to transfer features.

With the emergence of deep learning (DL), end-to-end feature extraction can be achieved based on DL methods, which can extract deeper and more comprehensive features. The existing DA methods based on DL are mainly divided into adversarial-based and discrepancy-based methods [13]. Adversarial-based methods aim to learn transferable features by adversarially training the generator and the discriminator [14]. Ganin et al. [15] introduced a domain adversarial neural network (DANN) that uses an adversarial strategy to align features between two domains. Tzeng et al. [16] used source samples for pretraining, and then used target samples for adversarial training to obtain the target encoder, which enabled the discriminator to distinguish the differences between the source and target domains. Huang et al. [17] designed a two-branch attention adversarial DA (TAADA) network, which uses a two-branch feature extraction subnetwork as a generator and two multilayer linear structure as a discriminator for distribution alignment. Discrepancy-based methods aim to align the marginal or conditional distributions between the source and target domains. Long et al. [18] proposed a deep adaptation network (DAN), which uses an optimal multikernel selection method for marginal distribution

Received 26 September 2024; revised 3 December 2024; accepted 6 December 2024. Date of publication 16 December 2024; date of current version 3 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42471417, Grant 42471404, and Grant 42171351; and in part by the National Key Research and Development Program of China under Grant 2020YFA0714200. *(Corresponding authors: Jie Xu; Jiangtao Peng.)*

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Digital Object Identifier 10.1109/TGRS.2024.3518502

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alignment. Zhu et al. [19] designed a multirepresentation adaptation network (MRAN) and an inception adaptation module (IAM) to align multiple representations. Inspired by subdomain adaptation, they further present a deep subdomain adaptation network (DSAN) to align subdomain distributions based on a local maximum mean discrepancy (LMMD) [20]. Fang et al. [21] combined DA with confident learning (CLDA) to reduce the discrepancy and generate pseudolabels with high confidence. Zhang et al. [22] used a graph neural network architecture to design a graph optimal transmission for aligning topological relationships between domains to minimize discrepancies (TSTNet). Cai et al. [23] used knowledge from three levels: image, feature, and logic to perform multilevel feature alignment [multilevel unsupervised domain adaptation (MLUDA)]. However, these DL-based methods mainly focus on the data distribution alignment problem and rarely consider the category characteristics of the data.

In recent years, contrastive learning has been widely used in computer vision [24], [25], natural language processing [26], [27], and other fields [28] and has also proven its effectiveness in learning domain-invariant features. Contrastive learning aims to learn domain-invariant features by maximizing (minimizing) the similarity of positive (negative) sample pairs, which uses contrastive loss to control similar (dissimilar) samples closer (farther). Huang et al. [29] designed a category contrast method to learn domain-invariant features using instance-to-instance contrastive learning. Jiang et al. [30] introduced an adaptation method based on prototypical contrastive learning, which aligns the distribution in a class-centered way. In HSI DA field, Ning et al. [31] proposed a category matching method combining contrastive learning to learn instance-level discriminative feature embeddings. On this basis, they further designed a prototype contrast method and proposed a bidirectional interdomain instance-to-prototype contrastive loss to extract compact discriminative features [32]. Li et al. [33] also proposed a supervised contrastive learning method to enhance the separability of intradomain data [supervised contrastive learning-based unsupervised domain adaptation (SCLUDA)]. All these contrastive learning-based methods need to construct positive and negative sample pairs, usually by adding Gaussian noise or random flip. However, such construction methods only consider single domain information or do not adequately consider the interaction of domain information, resulting in large differences in the learned domain-invariant features, which has a negative impact on knowledge migration.

To introduce more effective domain information for domain fusion, we consider combining mixup techniques in contrastive learning. Mixup is an effective image mixing technique that obtains an enhanced image by performing a pixel-weighted combination of two global images [34]. Many works in the field of computer vision have also demonstrated its effectiveness [35], [36], [37]. Tranheden et al. [38] mixed images from both the domains along with their corresponding labels and pseudolabels and achieved DA by training labeled and mixed samples simultaneously. He et al. [39] constructed a unified input space for the source and target domains through nonlocal mixup to align features and transfer knowledge. Kim et al. [40] introduced a bidirectional domain mixup method to handle DA tasks. Li et al. [41] proposed an interdomain mixup strategy, which mixup sample-level and feature-level data to facilitate cross-domain feature alignment. Eldele et al. [28] mixup the source and target domain samples to form an enhanced view, and then the similarity between each domain and its corresponding enhanced view is maximized through contrastive learning, thereby solving the distribution shift problem between domains. HSIs differ from traditional optical images in that they have hundreds of bands, often with redundant or noisy bands. If the mixup operation is performed directly at the image level, noise information will be introduced, which will have a negative impact on the migration effect. To introduce domain information while avoiding noise interference, we perform domain fusion at the feature level, introduce more refined domain features, and propose a domain fusion contrastive learning (DFCL) framework for cross-scene HSI classification.

In the proposed DFCL method, the features extracted by convolutional neural networks are more refined and more conducive to mixup contrast. To learn more sufficient domain-invariant features, we use inter- and intradomain fusion methods, to generate corresponding enhanced views. Interdomain fusion is to enable the model to learn discriminative features of the source domain and target domain in contrastive learning. Intradomain fusion is to maximize the similarity between samples of the same category and the distance between samples of different categories, so as to learn the distinctive features of each category of sample. To achieve a more similar distribution between the source and target domains, we fusion the source and target domains to obtain an intermediate domain and use feature similarity and label similarity to minimize the distance between the domains. In addition, maximum mean discrepancy (MMD) is also introduced to align the feature distribution between the source domain and the target domain. Finally, since the target domain samples are unlabeled in contrastive learning, to obtain pseudolabels with high confidence, a progressive strategy based on prototype learning is used to select more credible pseudolabels through the entropy value of the pseudolabels.

The contributions of our proposed DFCL can be summarized as follows.

1) A novel DFCL method is proposed. It can effectively use domain information to solve the domain transfer problem, which mainly includes inter- and intradomain fusion, which not only pulls the samples of the same category in the same domain closer together but also separates the similar features of the samples of different categories in different domains.
2) We introduce a feature space loss based on feature similarity and label similarity for measuring the difference between the intermediate domain and source (target) domain, thus minimizing the domain discrepancy.
3) We design a progressive label selection strategy based on prototype learning to obtain labels with high confidence in the target domain. These pseudolabels with high confidence are used for contrastive learning, which is more beneficial to learn the domain-invariant features, and have a positive impact on fusion contrastive learning.

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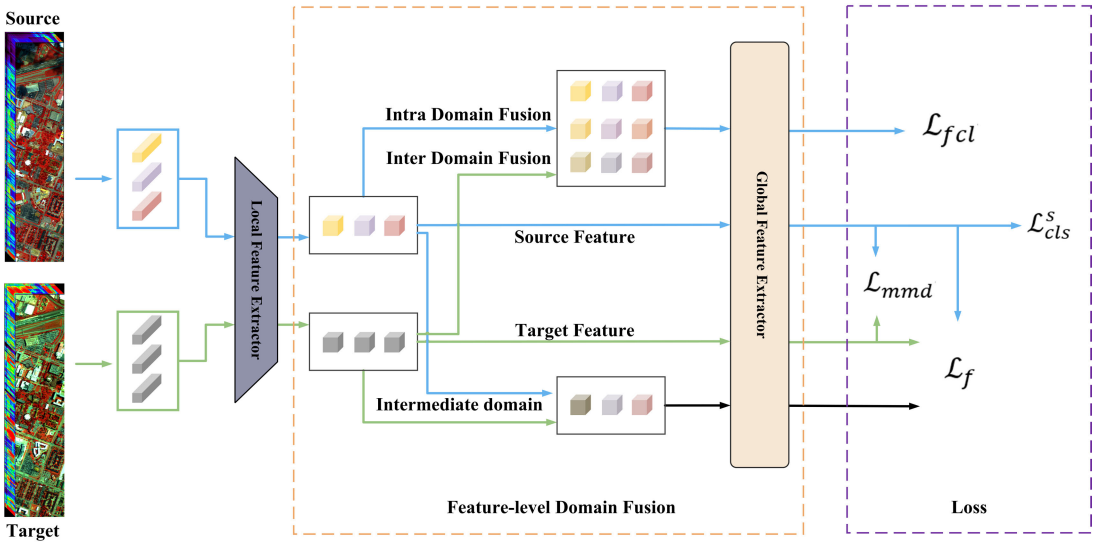


Fig. 1. Framework of the proposed DFCL. The blue and green lines show the computational flow of the source domain samples and target domain samples, respectively.

The remainder of the article is as follows. Section II introduces the details of DFCL method. Section III demonstrates the experimental results and analysis. Section IV draws the conclusion.

II. Proposed Method

For DA tasks, the source and target domains are of the same categories, while the source domain is labeled and the target domain without labels. Assuming that given a source domain $D_s = (X_s, Y_s)$ with n_s samples $\{(x_s^i, y_s^i)\}_{i=1}^{n_s}$ and a target domain $D_t = (X_t, Y_t)$ with n_t samples $\{x_t^i\}_{i=1}^{n_t}$, both the domains have C categories. DA aims to migrate the model obtained by training the labeled source domain to the target domain, thereby obtaining the sample labels of the target domain. The overall framework of the proposed method DFCL is shown in Fig. 1.

DFCL uses a mixup strategy for inter- and intradomain fusion at the feature level to augment the data and uses contrastive learning to obtain domain-invariant features. In addition, to reduce domain discrepancy, feature space loss is used to measure the distance between the intermediate domain features and the source (target) domain and MMD is used align source and target domain distributions. Finally, the pseudolabels of the target domain are obtained using a progressive strategy based on prototype learning for contrastive learning.

A. Feature Extractor Module

In the proposed DFCL method, the feature extraction module contains two parts: local feature extractor and global feature extractor. Fig. 2 shows the architecture of the local feature extractor, where features are extracted using 3-D convolution and 2-D convolution. In addition, considering the feature redundancy and noise in the spatial spectral information, we add spatial attention and spectral attention mechanisms. Both of them use maximum pooling and average pooling operations on features. The convolution kernel size used for spatial attention is 7×7, and the convolution kernel size used for spectral attention is 1×1. The global feature extractor uses the original transformer encoder architecture and exploits its multihead self-attention mechanism to extract global features.

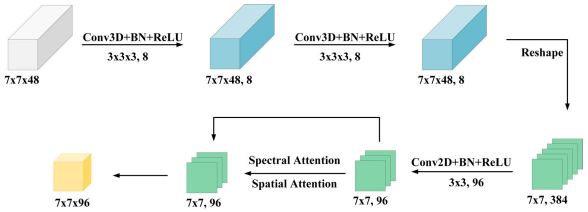


Fig. 2. Architecture of a local feature extractor.

B. Domain Fusion Contrastive Learning

Contrastive learning is designed to be a way to compare the views of the same category of samples with different enhancements, thus minimizing the discrepancy between samples of the same category and maximizing the discrepancy between samples of different categories. This is usually done by adding Gaussian noise or random flip, without taking domain information into account. Our proposed method performs domain fusion at the feature level, using both inter- and intradomain fusion to augment the samples. In the presence of domain information interference, more discriminative domain-invariant features will be learned by contrastive learning.

1) Interdomain Fusion: Mixup is an effective data augmentation mechanism that prevents model overfitting by interpolating two samples. Mixup types are categorized into global mixup and local mixup. Global mixup includes: Mixup [34] and Manifold-Mixup [42]. Local mixup includes CutMix [43], TransMix [44], and TokenMix [45]. Considering the characteristics of HSI and the utilization of mixup strategies at the feature level. We use the vanilla global mixup: Mixup. When mixup is performed, the features of the dominant domain need to be preserved while considering the information from the other domain. For interdomain fusion,

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when enhancing source domain samples, the proportion of source domain samples must be greater than that of target domain samples. Conversely, the proportion of target domain samples is larger. So mixup ratio λ_1 is randomly selected from (0.7, 1). Given N source domain features $\{a_s^i\}_{i=1}^N$ and N target domain features $\{a_t^i\}_{i=1}^N$ obtained by the local feature extractor, we generate the source domain augmented view a_{st} as follows:

$$a_{st}^i = \lambda_1 a_s^i + (1-\lambda_1) a_t^i. \tag{1}$$

Similarly, the same process is used to generate augmented views of the target domain a_{ts}, which is formalized as follows:

$$a_{ts}^i = (1-\lambda_1) a_s^i + \lambda_1 a_t^i. \tag{2}$$

2) Intradomain Fusion: We use the same mixup strategy as for intradomain fusion. The difference is that for intradomain fusion, we need to break up the order of the samples in the domain. For N source features $\{a_s^i\}_{i=1}^N$, randomly disrupt their order to generate the order inconsistent $\{\hat{a_s^i}\}_{i=1}^N$. In addition, since intradomain fusion also needs to ensure that the dominant category is preserved, the mixup ratio λ_2 is also chosen randomly from (0.7, 1). Formally, we use intradomain fusion to generate the source domain augmented view as follows:

$$a_{ss}^i = \lambda_2 a_s^i + (1-\lambda_2) \hat{a_s^i}. \tag{3}$$

Likewise, the process of generating an augmented view of the target domain is as follows:

$$a_{tt}^i = \lambda_2 a_t^i + (1-\lambda_2) \hat{a_t^i}. \tag{4}$$

3) Contrastive Learning: Inter- and intradomain fusion is performed at the feature level to obtain augmented views of the source and target domains, a_{st}, a_{ts}, a_{ss}, and a_{tt}, respectively. Taking the N source domain features as an example, $\{a_s^i\}_{i=1}^N$ and its corresponding augmented views are obtained as output features through the global feature extractor $\{z_s^i\}_{i=1}^N$, $\{z_{st}^i\}_{i=1}^N$, and $\{z_{ss}^i\}_{i=1}^N$. The corresponding fusion contrastive loss for the source domain is defined as

$$\mathcal{L}_{\text{fcl}}^{z_s^i} = -\frac{1}{\left|I_{z_s^i}^+\right|} \sum_{z_i^+ \in I_{z_s^i}^+} \log \frac{\phi\left(z_s^i, z_i^+\right)}{\phi\left(z_s^i, z_i^+\right) + \sum_{z_i^- \in I_{z_s^i}^-} \phi\left(z_s^i, z_i^-\right)} \tag{5}$$

where $\phi(m,n) = \exp(m \cdot n/\tau)$, and $I_{z_s^i}^+ (I_{z_s^i}^-)$ represent a set of positive (negative) matching samples. $|I_{z_s^i}^+|$ is the number of positive matching set. So for N samples, the final contrastive loss is defined as

$$\mathcal{L}_{\text{fcl}}^s = \frac{1}{N} \sum_{z_s^i \in \{z_s^i\}_{i=1}^N} \mathcal{L}_{\text{fcl}}^{z_s^i}. \tag{6}$$

The contrast loss for the target domain is defined as

$$\mathcal{L}_{\text{fcl}}^{z_t^i} = -\frac{1}{\left|I_{z_t^i}^+\right|} \sum_{z_i^+ \in I_{z_t^i}^+} \log \frac{\phi\left(z_t^i, z_i^+\right)}{\phi\left(z_t^i, z_i^+\right) + \sum_{z_i^- \in I_{z_t^i}^-} \phi\left(z_t^i, z_i^-\right)} \tag{7}$$

$$\mathcal{L}_{\text{fcl}}^t = \frac{1}{N} \sum_{z_t^i \in \{z_t^i\}_{i=1}^N} \mathcal{L}_{\text{fcl}}^{z_t^i} \tag{8}$$

where $I_{z_t^i}^+ (I_{z_t^i}^-)$ represent a set of positive (negative) matching samples. However, source domain samples have corresponding labels, while target domain samples do not. To obtain the labels of the target domain, we adopt a progressive generation strategy to generate the corresponding pseudolabels through prototype learning.

C. Domain Adaptation

In the DA task, it is important to minimize the discrepancies between the source and target domains and to solve the domain shift problem. We are inspired by Zhu et al. [46] and use cosine similarity and cross-entropy (CE) function to calculate the similarity between domains. Moreover, we use the MMD distance to align the domain distribution.

1) Feature Space Loss: We mixup the source and target domains to obtain an intermediate domain, which is used to measure the distance to the source and target domains. Unlike the previous mixup strategy, here the mixup ratio λ_3 is selected randomly from (0, 1). The intermediate domains are generated as follows:

$$a_{\text{inter}}^i = \lambda_3 a_s^i + (1-\lambda_3) a_t^i. \tag{9}$$

The output features $\{z_s^i\}_{i=1}^N$, $\{z_{\text{inter}}^i\}_{i=1}^N$, $\{z_t^i\}_{i=1}^N$ are obtained through global feature extractor. Take the source domain as an example, calculate the feature space loss between the source and intermediate domains. First, the feature similarity matrix M_s is obtained by cosine similarity calculation, and then the label similarity is calculated by $y_{\text{sl}} = y_s (y_s)^T$ to obtain a binary matrix. This matrix indicates whether the two samples belong to the same category or not, which is 1 if they belong to the same category and 0 otherwise. Finally, the CE function is used to measure the discrepancies between the domains by feature similarity and label similarity. The feature space loss for the source domain is defined as

$$M_s^{ij} = \frac{z_{\text{inter}}^i \cdot z_s^j}{||z_{\text{inter}}^i|| \times ||z_s^j||} \tag{10}$$

$$\mathcal{L}_f^s = -\frac{\lambda_3}{N} \sum_{i=1}^N y_{\text{sl}}^i \log M_s^i \tag{11}$$

where M_s^i is normalized to (0, 1) through softmax for calculation. Similarly, the feature space loss in the target domain is computed, and the overall loss is as follows:

$$\mathcal{L}_f = \mathcal{L}_f^s + \mathcal{L}_f^t. \tag{12}$$

2) MMD Loss: After the final source domain features and target domain features are obtained, the MMD loss is defined as follows:

$$\mathcal{L}_{\text{mmd}} = || \frac{1}{N} \sum_{i=1}^N \varphi\left(z_s^i\right) - \frac{1}{N} \sum_{j=1}^N \varphi\left(z_t^j\right) ||_{\mathcal{H}}^2 \tag{13}$$

where $\varphi(\cdot)$ is the Gaussian kernel function.

So the total DA loss is as follows:

$$\mathcal{L}_{\text{da}} = \mathcal{L}_f + \alpha \mathcal{L}_{\text{mmd}} \tag{14}$$

where $\mathcal{L}_f$ represents the feature space loss, $\mathcal{L}_{\text{mmd}}$ represents the MMD loss, and α is a weight parameter.

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Algorithm 1 DFCL

Input: Input source sample set $D_s = (X_s, Y_s)$; target sample set $D_t = X_t$; the weight parameters α.

Output: Output predicted labels of target domain.

for $i = 1$ to *epochs* **do**
  Perform local feature extractor to extract the local features to get a_s and a_t.
  Perform interdomain fusion by (1) and (2) to get source and target domain augmented views a_{st} and a_{ts}.
  Perform intradomain fusion by (3) and (4) to get source and target domain augmented views a_{ss} and a_{tt}.
  Generate intermediate domain by (9).
  Perform global feature extractor get output features z_s, z_t, z_{st}, z_{ts}, z_{ss}, z_{tt}, z_{inter}.
  Compute the prototype and obtain a pseudolabel $\tilde{y}_t$ for the target domain.
  Obtain contrastive loss by (5)-(8) and domain adaptation loss by (14).
  Calculate cross-entropy loss of source sample.
  Update the model parameters by (20).
end for
Obtain predicted labels Y_t of the target domain.

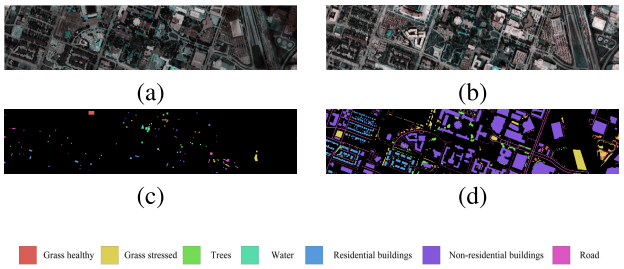


Fig. 3. Houston datasets. (a) False-color image of Houston13. (b) False-color image of Houston18. (c) Ground-truth map of Houston13. (d) Ground-truth map of Houston18.

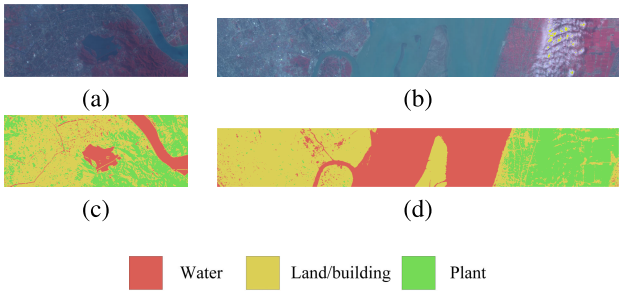


Fig. 4. Hangzhou–Shanghai datasets. (a) False-color image of Hangzhou. (b) False-color image of Shanghai. (c) Ground-truth map of Hangzhou. (d) Ground-truth map of Shanghai.

D. Progressive Prototype Learning

Previous work on DA has shown that prototype learning is effective in generating pseudolabels [47], [48]. To generate high-confidence labels, we convert the distance from the sample to the prototype into a probability distribution based on prototype learning, obtain the entropy value of the sample based on its probability distribution, and select the one with smaller entropy value as a high-confidence label.

The prototype of the kth class in D_s can be calculated as

$$c_k = \frac{1}{|\mathcal{Z}_k|} \sum_{z_i \in \mathcal{Z}_k} z_i \tag{15}$$

where $\mathcal{Z}_k$ is the sample set of the kth class in $\{z_s^i\}_{i=1}^N$.

Based on these prototypes, the pseudolabel of the ith target sample is obtained by measuring the distance between the samples to each class prototype

$$d_i = \text{distance}\left(c, z_t^i\right) \tag{16}$$

$$\tilde{y}_t^i = \text{argmin}(d_i) \tag{17}$$

where distance uses the Euclidean distance. Then we converted the distance from the sample to the prototype into a probability distribution and calculate its entropy value

$$p_i = \frac{\varphi(d_i)}{\sum_k \varphi(d_i)} \tag{18}$$

$$e_i = -\sum_{i=1}^{k} p_i \log(p_i) \tag{19}$$

where $\varphi(\cdot)$ is the Gaussian kernel function, p_i is the probability, and e_i is the entropy value of the ith target sample. During the training process, high-confidence labels are selected based on the ratio of the current iteration number to the total iteration number as a threshold value. For example, if the current number of iterations is 50 and the total number of iterations is 100, the top 50% of the sample labels with low entropy values are selected.

E. Total Loss

To improve model accuracy and stability, CE loss is applied to the source domain features for training the classifier. The total loss is as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}}^s + \mathcal{L}_{\text{fcl}}^s + \mathcal{L}_{\text{fcl}}^t + \mathcal{L}_{\text{da}}. \tag{20}$$

Algorithm 1 demonstrates the pseudocode of DFCL.

III. Experiments

A. Datasets

To validate the effectiveness of DFCL method, we conducted experiments on three datasets, including the Houston dataset, the Hangzhou–Shanghai dataset, and Yancheng dataset, respectively.

1) Houston: This dataset was acquired from Houston University and its surrounding area using different sensors in 2013 and 2018, respectively. The Houston2013 [49] and Houston2018 [50] datasets consisted of 349 × 1905 and 209 × 955 pixels, respectively. The former encompasses 144 bands with 15 distinct categories, while the latter includes 48 bands with 20 categories. To analyze the experimental results, 48 bands and seven categories were selected for both the datasets. Table I shows the specific information. Fig. 3 shows the false-color images and ground-turth maps of Houston dataset.

2) Hangzhou–Shanghai: This dataset was collected in Hangzhou and Shanghai using the EO-1 Hyperion hyperspectral sensor [51]. The two datasets consist of 590 × 230 and 1600 × 230 pixels, respectively, and both contain 198 bands and three common categories. The details are shown in

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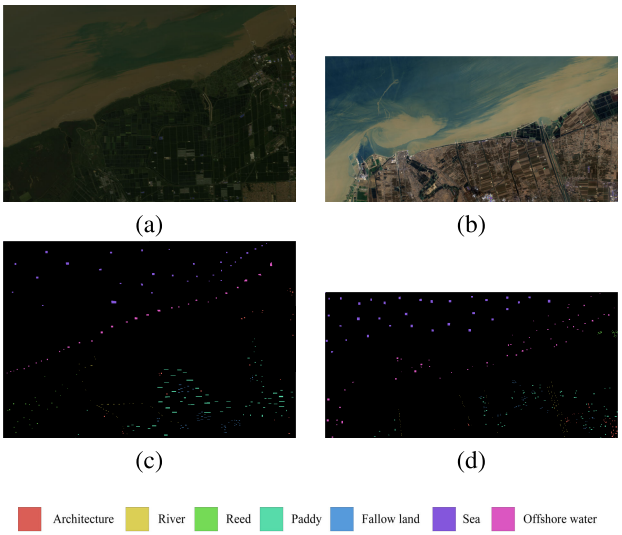


Fig. 5. Yancheng datasets. (a) False-color image of ZY-YC. (b) False-color image of GF-YC. (c) Ground-truth map of ZY-YC. (d) Ground-truth map of GF-YC.

TABLE I
NUMBER OF SAMPLES IN THE HOUSTON DATASET

No.	Class Name	Houston13 (Source)	Houston18 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

TABLE II
NUMBER OF SAMPLES IN THE HANGZHOU–SHANGHAI DATASET

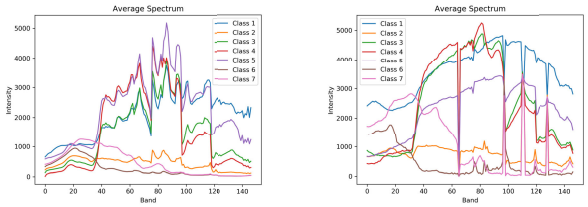
No.	Class Name	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/building	77450	161689
3	Plant	40207	83188
	Total	135700	368000

Table II. Fig. 4 shows the false-color images and ground-turth maps of Hangzhou-Shanghai dataset.

3) Yancheng: This dataset is provided by the Advanced Hyperspectral Imager on board China's Zhiyuan 1-02D (ZY1-02D) and Gaofen 5 (GF-5) satellites [52]. ZY1-02D Yancheng (ZY-YC) includes 1398 × 942 pixels, while GF-5 Yancheng (GF-YC) includes 1175 × 585 pixels. Both the datasets contain 147 bands and seven categories. The details are shown in Table III. Fig. 5 shows the false-color images and ground-turth maps of Hangzhou-Shanghai dataset. In addition, Fig. 6 shows the average spectrum of the Yancheng dataset. It is obvious that the distributions of the source and target domains are different, and the target domain also contains noise bands, making classification more challenging.

TABLE III
NUMBER OF SAMPLES IN THE YANCHENG DATASET

No.	Class Name	ZY-YC (Source)	GF-YC (Target)
1	Architecture	451	360
2	River	214	217
3	Reed	244	132
4	Paddy	3026	832
5	Fallow land	650	234
6	Sea	2076	2395
7	Offshore water	1558	1305
	Total	8219	5475



(a) (b)

Fig. 6. Mean spectrum of all classes on the Yancheng dataset. (a) Average spectrum image of ZY-YC. (b) Average spectrum image of GF-YC.

B. Experimental Setups

We select the following DL-based DA methods to compare with the proposed DFCL method, i.e., DAN [18], DSAN [20], MRAN [19], CLDA [21], TSTNet [22], SCLUDA [33], and MLUDA [23]. In the experiments, with respect to the size of the patch size, it is 7 × 7 on the Houston and Yancheng datasets, and 3 × 3 on the Hangzhou–Shanghai datasets. The batch size and the number of iterations are set to 64 and 100, respectively. The initial learning rates on the three datasets are 0.001, 0.0003, and 0.0001, respectively. For the parameter τ of SCLUDA and the proposed DFCL is set to 0.1. In addition, for DAN, we use ResNet-50 as the feature extractor, while for DSAN and MRAN, the spectral–spatial dual-branch attention convolutional network is used as the feature extractor.

To evaluate the results of each method, overall accuracy (OA), average accuracy (AA), and κ coefficient are used for verification purposes.

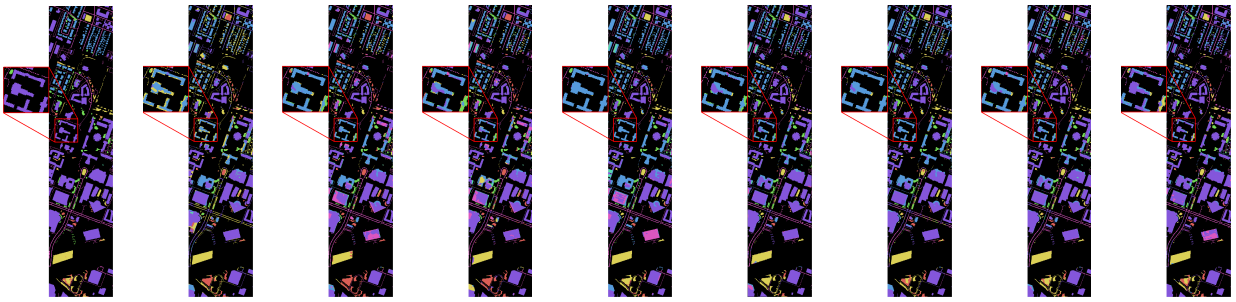
C. Comparison Experiments

1) Experiments on the Houston Dataset: The Houston dataset is a more challenging dataset as it was collected at different times and by different sensors. Table IV illustrates classification results of different methods at Houston2018 and the proposed DFCL performs better compared with other methods. The DAN method was the least effective at 55.19%. This is due to the inferior performance of the ResNet-50-based feature extractor in extracting HSI features, which is not well-suited for the HSI DA task. DSAN and MRAN use a HSI-specific infrastructure combined with a DA approach to achieve better results, reaching about 67%. For methods CLDA, TSTNet, SCLUDA, and MLUDA not only use HSI-specific feature extractors but also align feature distributions from different perspectives, so that superior

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TABLE IV
CLASSIFICATION RESULTS ON THE HOUSTON2018 DATASET

Class	DAN	DSAN	MRAN	CLDA	TSTNet	SCLUDA	MLUDA	DFCL
1	63.93	**96.48**	85.82	62.60	57.86	80.03	67.83	66.65
2	54.50	39.23	58.57	**88.75**	71.40	69.75	76.60	75.71
3	**66.11**	64.46	62.73	62.82	50.31	64.13	58.38	63.53
4	80.30	84.85	79.09	72.73	84.09	92.61	86.82	**93.18**
5	66.92	86.21	77.27	92.27	**92.81**	89.38	80.63	78.67
6	52.84	63.85	70.46	60.88	74.04	73.56	82.40	**84.18**
7	51.11	75.80	47.43	**79.71**	58.91	57.35	52.68	66.59
OA(%)	55.19 ± 1.19	66.14 ± 1.65	67.29 ± 1.86	68.99 ± 1.84	72.21 ± 1.87	72.56 ± 1.84	76.49 ± 2.70	**79.23 ± 1.60**
AA(%)	62.24 ± 4.34	72.98 ± 1.58	68.77 ± 3.13	74.25 ± 3.75	69.92 ± 2.36	75.37 ± 2.43	72.19 ± 5.36	**75.50 ± 0.74**
κ(%)	38.45 ± 0.94	51.98 ± 1.86	50.58 ± 2.58	56.29 ± 1.86	57.43 ± 3.37	58.41 ± 2.76	62.01 ± 4.63	**66.17 ± 1.75**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 7. Classification maps of the Houston2018 dataset. (a) Ground truth. (b) DAN. (c) DSAN. (d) MRAN. (e) CLDA. (f) TSTNet. (g) SCLUDA. (h) MLUDA. (i) DFCL.

TABLE V
CLASSIFICATION RESULTS ON THE SHANGHAI DATASET

Class	DAN	DSAN	MRAN	CLDA	TSTNet	SCLUDA	MLUDA	DFCL
1	87.95	91.51	94.93	93.57	93.87	94.21	94.88	**95.92**
2	81.50	87.15	76.11	86.01	84.36	84.63	85.21	**88.23**
3	89.25	82.74	99.62	99.60	98.96	99.45	**99.85**	99.03
OA(%)	85.52 ± 1.66	88.45 ± 1.37	87.50 ± 1.46	91.61 ± 1.72	90.84 ± 0.83	91.19 ± 0.87	91.76 ± 1.88	**93.24 ± 1.41**
AA(%)	86.23 ± 2.61	87.13 ± 3.71	90.22 ± 1.04	93.06 ± 1.34	92.40 ± 0.77	92.76 ± 0.74	93.32 ± 1.47	**94.39 ± 0.88**
κ(%)	77.93 ± 2.69	82.10 ± 2.09	81.21 ± 2.09	87.19 ± 2.57	86.04 ± 1.25	86.56 ± 1.31	84.44 ± 2.80	**89.66 ± 2.08**

TABLE VI
CLASSIFICATION RESULTS ON THE GF-YC DATASET

Class	DAN	DSAN	MRAN	CLDA	TSTNet	SCLUDA	MLUDA	DFCL
1	46.87	50.43	38.75	55.93	**66.93**	52.83	65.99	47.77
2	29.73	60.69	40.76	**98.46**	79.72	24.70	65.16	87.67
3	49.62	20.24	24.24	5.56	34.35	11.29	26.81	**56.44**
4	88.32	**99.64**	96.63	99.48	79.93	97.07	85.79	84.68
5	13.78	4.42	18.16	0	4.13	24.32	21.20	**58.76**
6	90.44	87.54	90.10	87.61	97.48	99.69	98.16	**100**
7	100	100	99.52	99.87	98.53	**99.95**	99.03	99.85
OA(%)	82.61 ± 2.61	83.61 ± 3.61	83.31 ± 2.83	84.96 ± 2.65	86.79 ± 1.96	87.95 ± 1.86	88.00 ± 2.82	**90.90 ± 2.32**
AA(%)	59.82 ± 3.45	60.71 ± 3.70	58.31 ± 2.92	63.84 ± 4.26	65.87 ± 5.75	58.55 ± 5.17	66.02 ± 2.64	**76.45 ± 3.81**
κ(%)	76.67 ± 3.42	77.98 ± 4.22	77.44 ± 3.40	79.71 ± 3.47	81.84 ± 2.64	83.20 ± 2.55	83.49 ± 3.78	**87.42 ± 3.16**

results are achieved. For example, SCLUDA uses contrastive learning to separate data in the domain and uses LMMD align distribution of subdomains. MLUDA uses multilevel alignment for knowledge migration. The proposed DFCL takes into account feature-level domain information and uses it for contrastive learning to learn more discriminative features, achieving optimal results with 79.23%.

Fig. 7 demonstrates the classification map of each method on the Houston2018 dataset. To better compare the effects between the different methods, we zoom in on the area of Class 6. It is apparent that the classification map of DFCL is closer to the ground truth. Classes 5 and 6 are "Residential buildings" and "Nonresidential buildings," respectively, which have similar spectral features, so all the methods misclassify Class 6 as Class 5. Especially for the CLDA methods, since the high confidence of the pseudolabel of Class 5 during the confidence learning process, Class 6 will basically be misclassified as Class 5. The proposed DFCL uses inter- and intradomain information for contrastive learning, which can more accurately discriminate between similar categories.

2) Experiments on the Hangzhou–Shanghai Dataset: Table V presents classification results of different methods at

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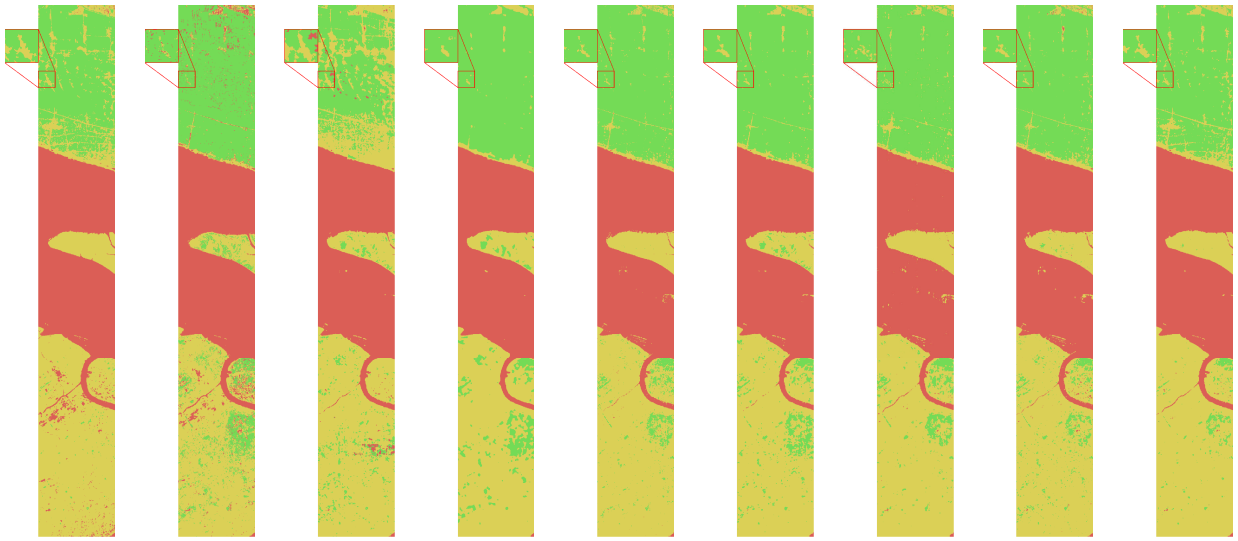


Fig. 8. Classification maps of the Shanghai dataset. (a) Ground truth. (b) DAN. (c) DSAN. (d) MRAN. (e) CLDA. (f) TSTNet. (g) SCLUDA. (h) MLUDA. (i) DFCL.

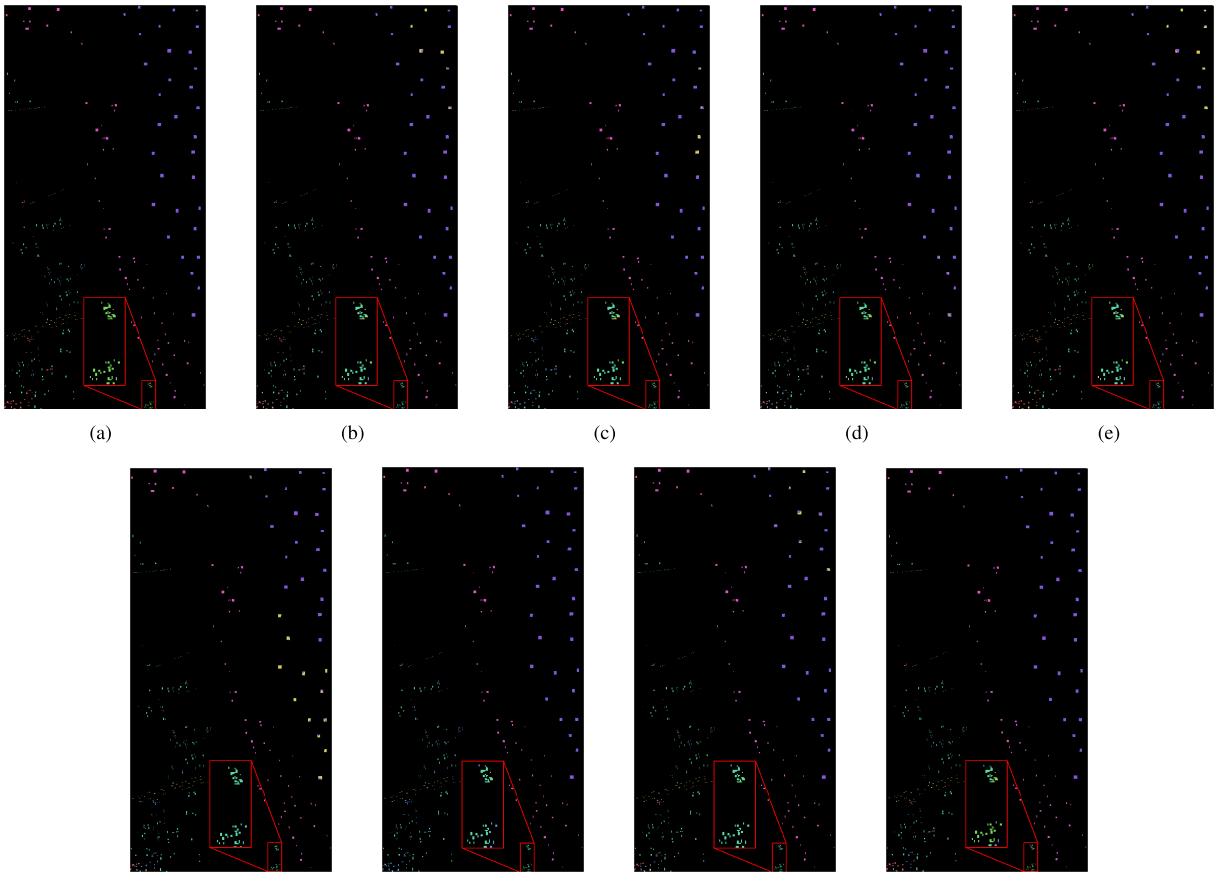


Fig. 9. Classification maps of the GF-YC dataset. (a) Ground truth. (b) DAN. (c) DSAN. (d) MRAN. (e) CLDA. (f) TSTNet. (g) SCLUDA. (h) MLUDA. (i) DFCL.

Shanghai. Similar to the Houston dataset, the performance of DAN on the Shanghai dataset is also the worst at only 85.52%, much lower than other DA methods. The classification results of DSAN and MRAN methods reach about 88%. The CLDA, TSTNet, SCLUDA, and MLUDA achieve about 91% of the classification results. The proposed DFCL achieves 93.24% classification result, which is 7.72% and 1.48% higher than DAN and MLUDA, respectively. For AA and κ, DFCL also achieves excellent results, with 94.39% and 89.66%, respectively. In particular, the value of κ, with CLDA differing by 2.47%. In addition, DFCL also achieves the best results in Classes 1 and 2.

To better demonstrate the classification results, we display the classification map of each method in Fig. 8. We zoom in

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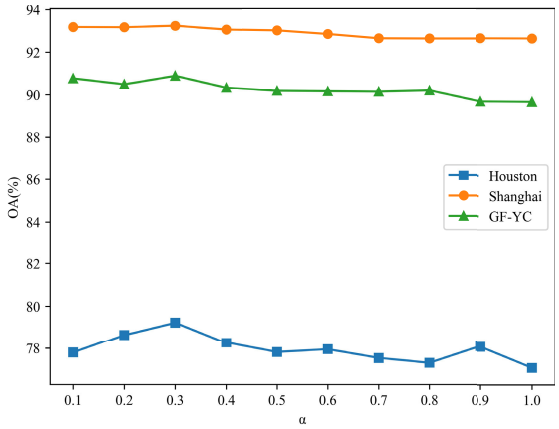


Fig. 10. Parameter α analysis.

a part of the area close to Classes 2 and 3. Obviously, the classification map of DFCL is closest to the ground truth. For DAN and DSAN, Class 1 appears in the classification map. For the other methods, it is a misclassification of Class 2 into Class 3. In contrast, our method has the lowest misclassification rate of the other methods.

3) Experiments on the Yancheng Dataset: For the Yancheng dataset, it is also a difficult classification task because of the variations in ground objects and different sensors used. Table VI displays the classification results of each method. The DAN, DSAN, and MRAN consider only a single data alignment approach, which have the lowest efficiency, only about 83%. Especially, for Class 5, all of them are less than 20%. For CLDA, although the OA is relatively superior at 84.96%, the accuracy of Classes 3 and 5 is extremely inferior at 5.56% and 0%, respectively. This is due to the overconfidence in pseudolabels' generation. Although TSTNet, SCLUDA, and MLUDA achieve classification accuracies of 86.79%, 87.95%, and 88.00%, respectively, they are still inferior on Classes 3 and 5. The proposed DFCL not only has superior OA than other methods but also has much higher classification accuracy on Classes 3 and 5 than other methods. In addition, AA and κ are also much higher than other methods, reaching 76.45% and 87.42%, which also proves that our method is optimal.

We illustrate the classification map of GF-YC in Fig. 9 and zoom in on the area around Class 3. Classes 3 and 4 are "Reed" and "Paddy," respectively, both of which belong to the gramineae family, so they have similar spectral features. Therefore, it is difficult to distinguish between these two categories of samples. The proposed DFCL maximizes the discrepancy between samples of different categories through DFCL, which can better distinguish samples of different categories with similar features. As a result, other methods incorrectly classified Class 3 as Class 4, and only our method has a lower misclassification rate and identifies more Class 3.

D. Ablation Analysis

To further verify the effectiveness of all modules of the DFCL method, we conducted corresponding ablation experiments on three datasets as shown in Table VII. Experiments 1–6 conduct ablation experiments on three modules, including the DFCL module, the DA module, and the pseudolabels module. When the pseudolabels module is not used, the classification head is directly used to classify the target domain features to obtain the pseudolabels. Obviously, on the three datasets, only using a single CE loss has the worst effect. This is because the source domain knowledge cannot be effectively transferred to the target domain. When the DFCL module is used, the network can learn more discriminative domain-invariant features, so the classification accuracy on the three datasets is greatly improved. On this basis, the MMD loss and feature space loss are added to reduce the difference between domains, and the classification effect is further improved. Compared with the pseudolabels generated directly using the classification head, the pseudo-labelsobtained using the pseudolabels module have higher confidence, which reduces the negative impact when learning domain-invariant features. Therefore, when all three modules are used, our method achieves the best classification effect.

TABLE VII
OA (%) UNDER DIFFERENT MODULES IN THREE DATASETS

	$\mathcal{L}_{cls}$	$\mathcal{L}_{fcl}$	$\mathcal{L}_{mmd}$	$\mathcal{L}_{fsl}$	Label	Houston	Shanghai	GF-YC
1	✓					67.98	83.04	78.94
2	✓	✓				71.64	87.21	83.54
3	✓	✓	✓			72.01	88.36	84.92
4	✓	✓		✓		72.77	87.79	84.23
5	✓	✓	✓	✓		73.09	90.93	85.87
6	✓	✓	✓	✓	✓	**79.23**	**93.34**	**90.90**

TABLE VIII
OA (%) OF DIFFERENT PATCH SIZES

	3 × 3	5 × 5	7 × 7	9 × 9	11 × 11
Houston	74.00	72.32	**79.23**	74.49	71.57
Shanghai	**93.24**	92.53	91.67	91.11	90.87
GF-YC	85.25	90.43	**90.90**	89.14	87.66

E. Parameter Analysis

In the DA module, weight parameter α is used to assess the impact of each loss on the classification effect. To select the most appropriate weight, we choose from $[0.1, 1]$ and experiment on three datasets to observe their trend. In Fig. 10, it is obvious that the fluctuation of OA is more obvious on the Houston dataset. When $\alpha = 0.3$, the best result is obtained. On the Shanghai and GF-YC datasets, the fluctuation of OA is more stable. To ensure parameter consistency as much as possible, we choose $\alpha = 0.3$.

Different patch sizes contain different spatial spectral information. We experimented with five sizes of 3×3, 5×5, 7×7, 9×9, and 11×11 on three datasets. As shown in Table VIII, for the Houston and GF-YC datasets, 7×7 is the optimal patch size, while for the Shanghai dataset, the best patch size is 3×3.

F. Feature Visualization

To further verify the degree of sample aggregation of the proposed DFCL method, we use t-distributed stochastic neighbor embedding (tSNE) to visualize the feature distribution. As shown in Fig. 11, we select Classes 2, 3, and 6 in the Houston dataset and select the same number of source and target domain samples to visualize their original feature

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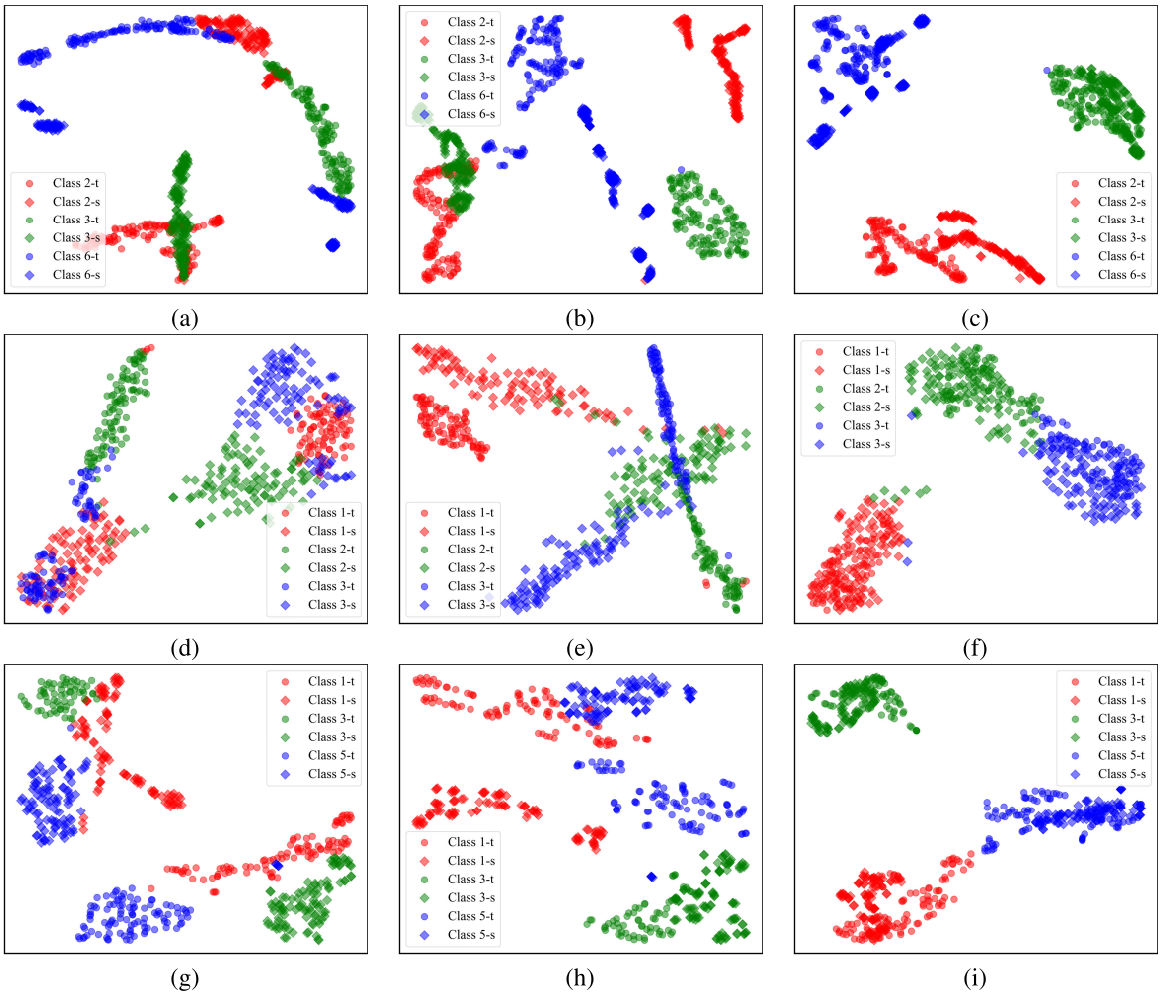


Fig. 11. Feature visualization on three datasets. Diamonds (circles) represent source (target) domain samples. (a) Original distribution for the Houston dataset. (b) SCLUDA distribution for the Houston dataset. (c) DFCL distribution for the Houston dataset. (d) Original distribution for the Hangzhou–Shanghai dataset. (e) SCLUDA distribution for the Hangzhou–Shanghai dataset. (f) DFCL distribution for the Hangzhou–Shanghai dataset. (g) Original distribution for the Yancheng dataset. (h) SCLUDA distribution for the Yancheng dataset. (i) DFCL distribution for the Yancheng dataset.

distribution and the features' distribution after SCLUDA and DFCL. For samples of the same category, the distribution in the source and target domains is relatively far apart. It is worth noting that some samples of Class 6 in the source domain are also distributed far away. Some samples of Class 3 in the source domain and Class 2 in the target domain have almost the same distribution. The SCLUDA method achieves good clustering effects on Classes 3 and 6 in the target domain, but Class 3 in the source domain and Class 2 in the target domain are not completely separated, which is because it only takes into account single-domain information in contrastive learning, and there are still differences in the learned domain-invariant features. Similarly, on the Hangzhou–Shanghai dataset, SCLUDA has a good clustering effect on Class 1, but does not separate Classes 2 and Class 3, while DFCL achieves better clustering effects on all three categories, with only a small number of samples overlapping between Class 2 and Class 3. For the Yancheng dataset, DFCL also has a better clustering effect than SCLUDA, especially in Classes 1 and 5. It can be clearly seen that the DFCL method brings samples of the same category in the source domain and the target domain closer together, while samples of different categories are pulled further apart. Therefore, the DFCL method can better reduce the discrepancies between domains and learn domain-invariant features.

TABLE IX
ONE EPOCH TRAINING TIME (S) FOR DIFFERENT METHODS

methods	DAN	DSAN	MRAN	CLDA	TSTNet	SCLUDA	MLUDA	DFCL
time	6.72	1.46	2.09	1.57	6.83	5.43	3.75	5.23

G. Analysis of the Computational Complexity

To compare the computational costs between different methods, we compare the training time on one epoch on the Houston dataset. As shown in Table IX, although DSAN, MRAN, and CLDA have short training times, their classification accuracy is far from satisfactory. For DAN and TSTNet, they use ResNet-50 and VGG-16 as feature extractors, so they have longer training times. For SCLUDA, most of its training time is spent on generating pseudo labels for the target domain. Although DFCL is not optimal in terms of time, it has higher classification accuracy than other methods.

IV. CONCLUSION

In this work, a DFCL framework has been proposed for cross-scene HSIs' classification. In using contrastive learning

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to solve the DA problems, different domain information needs to be combined as much as possible to learn more discriminative domain-invariant features. For target domain samples, the credibility of their corresponding pseudolabels is also extremely important. In DFCL, we use inter- and intradomain fusion to generate enhanced views, which fully considers the different domain knowledge, so that the learned domain-invariant features are more discriminative and similar features of different categories are separated in different domains. Moreover, we also design a feature space loss based on feature similarity and label similarity to measure the distance between domains, which can reduce effectively the difference between domains. Finally, regarding how to select high-confidence pseudolabels, we propose a progressive selection strategy based on prototype learning for generating pseudolabels, which has a positive effect on DFCL. The results on three datasets also prove the effectiveness of DFCL and that it outperforms other DA methods.

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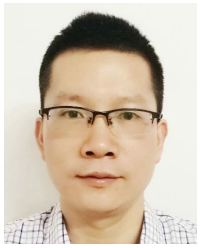
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